

$$V(a,b,c) = 11a^2 + 10b^2 + 13c^2 - 4ab - 6ac - 20bc$$

$$V(a,b,c) = (a \ b \ c) A \begin{pmatrix} a \\ b \\ c \end{pmatrix} \quad (a \ b \ c) \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

diagonala $a^2 \rightarrow 11$

$b^2 \rightarrow 10$

$c^2 \rightarrow 13$

$$A = \begin{pmatrix} 11 & -2 & -3 \\ -2 & 10 & -10 \\ -3 & -10 & 13 \end{pmatrix}$$

$$= \begin{pmatrix} a & b & c \end{pmatrix} \begin{pmatrix} A_{11}a + A_{12}b + A_{13}c \\ A_{12}a + A_{22}b + A_{23}c \\ A_{13}a + A_{23}b + A_{33}c \end{pmatrix}$$

$$= a(A_{11}a + A_{12}b + A_{13}c) + b(A_{12}a + A_{22}b + A_{23}c) + c(A_{13}a + A_{23}b + A_{33}c) =$$

$$= a^2 A_{11} + ab A_{12} + ac A_{13} + ab A_{12} + b^2 A_{22} + bc A_{23} + ac A_{13} + bc A_{23} + c^2 A_{33}$$

$$= a^2 A_{11} + b^2 A_{22} + c^2 A_{33} + 2ab A_{12} + 2ac A_{13} + 2bc A_{23}$$

$$A_{12} = \frac{-4}{2} = -2$$

$$A_{13} = \frac{-6}{2} = -3$$

$$A_{23} = \frac{-20}{2} = -10$$

Charakteristický polynom

$$\det \begin{pmatrix} 11-\lambda & -2 & -3 \\ -2 & 10-\lambda & -10 \\ -3 & -10 & 13-\lambda \end{pmatrix} = 0$$

$$-\lambda^3 + 34\lambda^2 - 339\lambda + 68 = 0$$

$$\lambda_1 \approx 0,22 \quad \lambda_2 \approx 10,5 \quad \lambda_3 \approx 23,3$$

$\lambda_1, \lambda_2, \lambda_3 > 0 \rightarrow A$ je pozitivně definitní $\rightarrow$
 proto platí $V(a,b,c) > 0$ pro $(a,b,c) \neq 0$

$$V(a,b,c) \geq 0 \text{ pro } a,b,c \in \mathbb{R}$$